

DevGuardian

AI-Powered Intelligent CI/CD Guardian

Microsoft Hackathon - Theme 06: AI-Powered Production Function: Reinventing Work

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Developer Trust Score + Codebase DNA Fingerprint

Zero-cost stack: NVIDIA NIM free models + GitHub APIs + open-source tooling

Updated build - 2026-06-13

1. Project overview

DevGuardian is an AI-native quality gate that plugs into GitHub pull-request workflows via a signed webhook. Unlike generic AI code reviewers, it adds two innovations: a Developer Trust Score (DTS) that routes each PR to the correct depth of AI scrutiny, and a Codebase DNA Fingerprint that learns each team's actual patterns and flags violations of THEIR standards - not generic rules. The whole system is a single FastAPI service that also serves a Tailwind dashboard SPA.

Theme	06 - AI-Powered Production Function: Reinventing Work
Core problem	PRs wait days for review; security bugs reach production; no intelligence between 'ready' and 'deploy'.
Unique angle	Developer Trust Score + Codebase DNA - stateful, per-developer intelligence that learns over time.
Tech cost	Rs 0 - NVIDIA NIM free-tier models + open-source tools + GitHub free APIs.
Backend	Python + FastAPI (single deploy; Docker-ready).
Storage	SQLite (trust + findings) + ChromaDB (DNA vector store).

2. The two core innovations

Developer Trust Score (DTS)

Every developer gets a 0-100 score recalculated from five signals. It is NEVER shown to managers - it is used only by the AI pipeline to decide review depth: high trust = faster feedback, low trust = deeper protection.

Bug introduction rate	bugs per 10 merged PRs	-20 pts
PR revert rate	% merged PRs reverted within 7 days	-15 pts
Test coverage delta	avg % coverage change	+0.3/%
Security findings	HIGH/CRITICAL findings (90 days)	-10 pts
First-pass accept rate	% PRs approved with no changes	+5/%

Formula: $DTS = 70 \text{ (baseline)} - \text{bug_rate} * 20 - \text{revert_rate} * 15 + \text{coverage_delta} * 0.3 - \text{security_count} * 10 + \text{accept_rate} * 5$, clamped to 0-100. Every score ships with an explainable factor-by-factor breakdown.

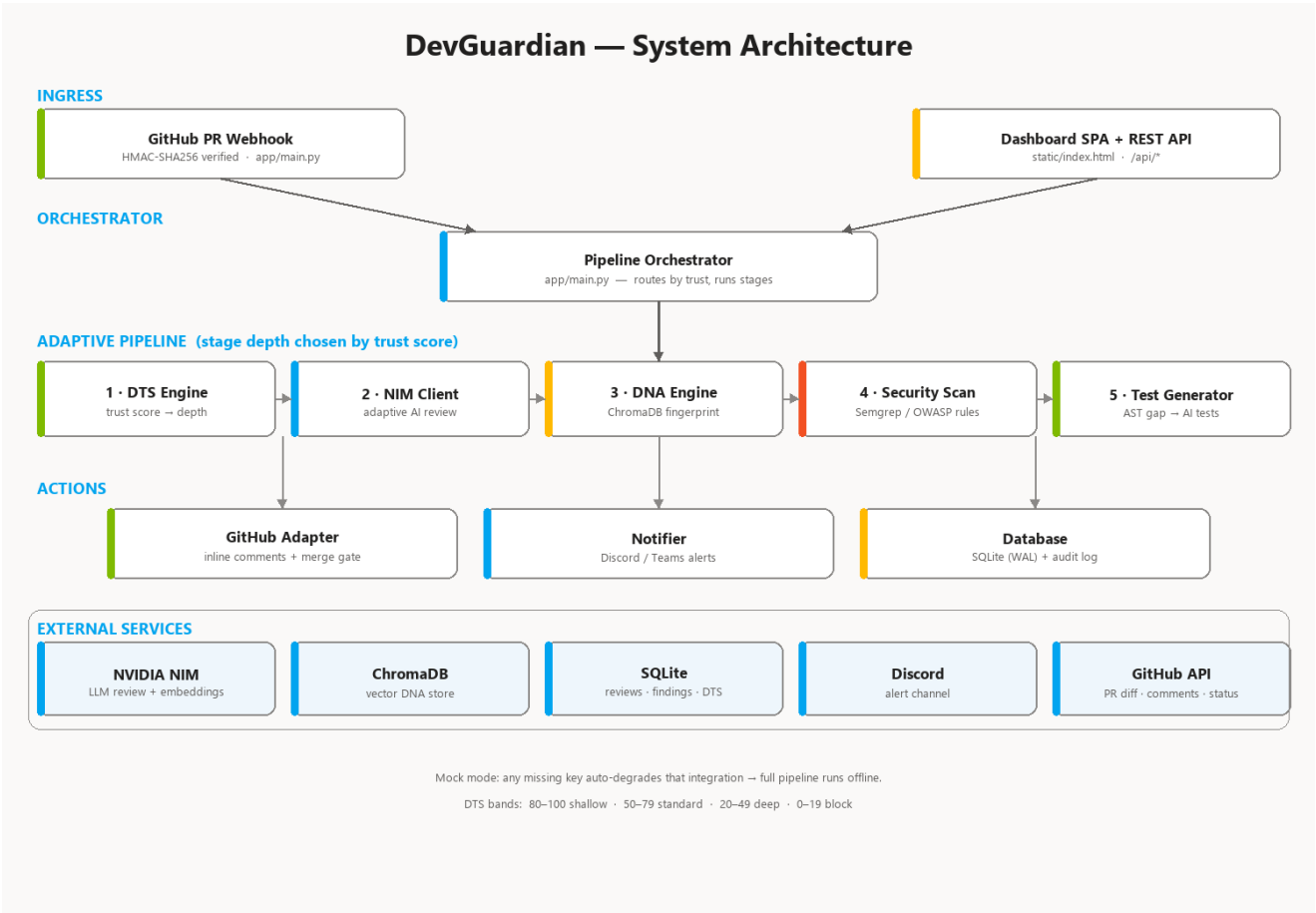
Codebase DNA Fingerprint

Merged code is embedded into ChromaDB (per repository). A new PR chunk whose nearest historical neighbours are too dissimilar is flagged as 'foreign DNA', and measurable conventions (naming, type hints, docstrings, logging, error handling) are compared against the learned profile - so every violation is explainable.

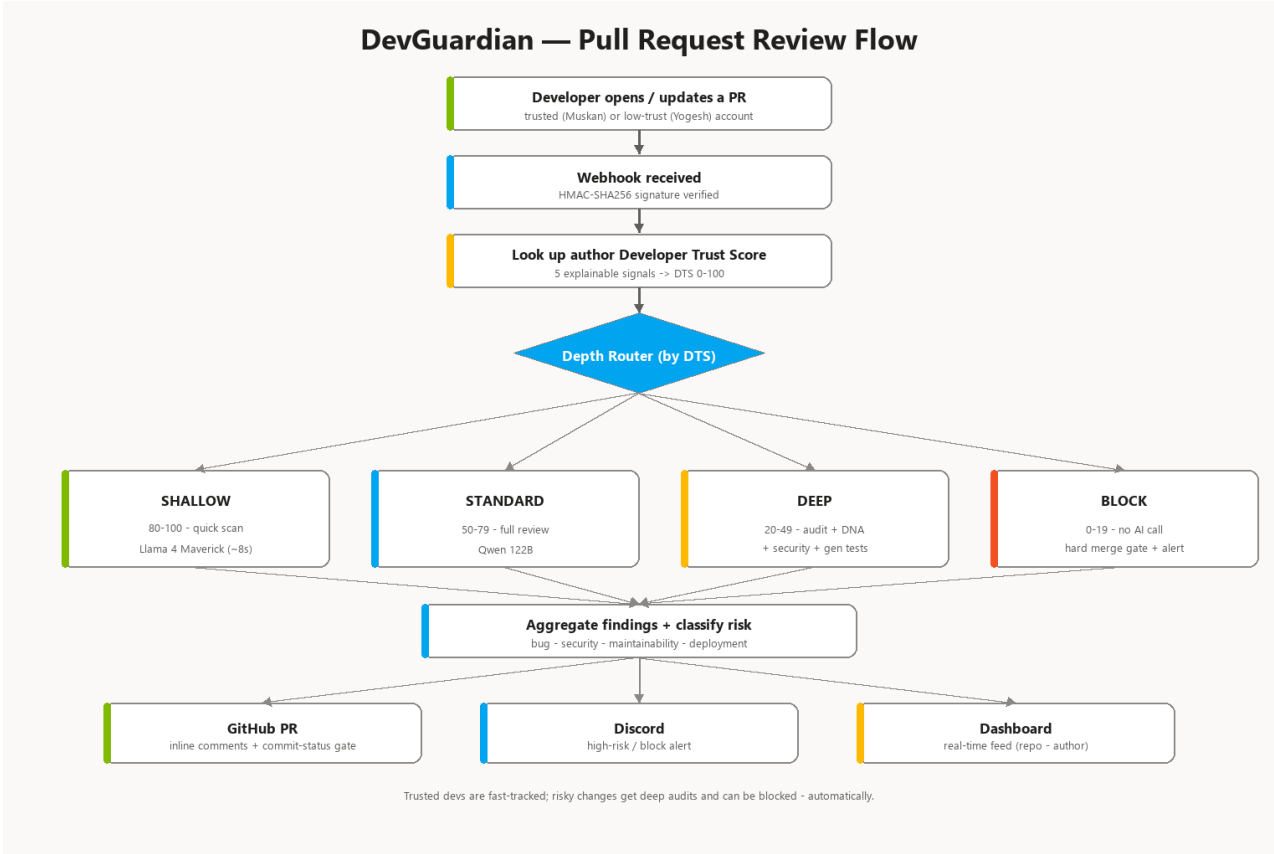
3. DTS to review-depth routing

80-100 Shallow	quick scan - meta/llama-4-maverick-17b (~8s)
50-79 Standard	full review - qwen/qwen3.5-122b-a10b
20-49 Deep	audit + DNA + security + generated tests - qwen/qwen3.5-122b
0-19 Block	no AI call - hard merge gate + team-lead alert
Fallback	nvidia/llama-3.3-nemotron-super-49b (on rate limit)
Embeddings	nvidia/nv-embedqa-e5-v5 (DNA)

4. System architecture



5. Pull-request review flow



6. What changed from the original brief

- Source control: GitHub PR webhooks (HMAC-verified) instead of Azure DevOps polling.
- Models updated to the live NVIDIA NIM catalogue (Llama 4 Maverick, Qwen3.5-122B, Nemotron-Super-49B fallback, nv-embedqa-e5-v5).
- Single FastAPI deploy serving a Tailwind SPA dashboard (no separate frontend).
- Multi-repo: per-repository DNA profiles + a repo filter on the dashboard.
- Real-time dashboard feed (auto-refresh) tagging every review and finding with its repo + author.
- Mock mode: any missing key auto-degrades that integration so the full pipeline runs offline.
- Resilient pipeline: a GitHub write-back failure never discards a completed review.